

Healthcare Analytics Dashboard Project Report

This report presents the complete workflow and analytical process followed for the Healthcare Analytics Dashboard project. The project involved data cleaning, transformation, analysis, visualization, and dashboard creation using Microsoft Excel.

Project Objectives

- Clean and preprocess healthcare-related data.
- Handle missing and inconsistent values.
- Transform raw columns into analytical features.
- Perform exploratory data analysis using charts and pivot tables.
- Create an interactive dashboard using slicers and KPI cards.
- Identify relationships between health conditions and healthcare charges.

Dataset Overview

The workbook contains the following sheets/tables: Customer Names, Medical Examinations, Hospitalisation Details, PivotTables, Dashboard, Healthcare. The dataset includes customer demographics, hospitalization details, medical examinations, BMI, HbA1C, surgeries, smoking status, and healthcare charges.

Sheet Name	Rows	Columns
Customer Names	2335	13
Medical Examinations	2335	10
Hospitalisation Details	2337	13
PivotTables	59	15
Dashboard	0	0
Healthcare	2335	17

Data Cleaning Process

- Checked missing values marked with '?' in Medical Examinations and Hospitalization Details tables.
- Filled missing month values with 'Sep'.
- Replaced missing year values using rounded average year (Formula : =ROUND(AVERAGE(B2:B2338),0).
- Filled missing smoker, hospital tier, and city tier values using mode/imputation.
- Handled missing State ID values using 'Unknown'.
- Verified data consistency in smoker and Heart Issues columns.

Customer ID	Year	Month	Days	Children	Charges	Hospital tier	City tier	State ID	Date of Birth	Age
Id2335	1992	Jul	9	0	\$563.84	tier - 2	tier - 3	R1013	09 July 1992	30
Id2334	1992	Nov	30	0	\$570.62	tier - 2	tier - 1	R1013	30 November 1992	30
Id2333	1993	Jun	30	0	\$600.00	tier - 2	tier - 1	R1013	30 June 1993	29
Id2332	1992	Sep	13	0	\$604.54	tier - 3	tier - 3	R1013	13 September 1992	30
Id2331	1998	Jul	27	0	\$637.26	tier - 3	tier - 3	R1013	27 July 1998	24
Id2330	2001	Nov	20	0	\$646.14	tier - 3	tier - 3	R1012	20 November 2001	21
Id2329	1993	Jun	1	0	\$650.00	tier - 3	tier - 3	R1013	01 June 1993	30
Id2328	1995	Jul	4	0	\$650.00	tier - 3	tier - 3	R1013	04 July 1995	27
Id2327	2002	Nov	29	0	\$668.00	tier - 3	tier - 2	R1012	29 November 2002	20
Id2326	1997	Nov	9	0	\$670.00	tier - 3	tier - 3	R1013	09 November 1997	25
Id2325	2001	Sep	12	0	\$687.54	tier - 3	tier - 2	R1013	12 September 2001	21
Id2324	1999	Dec	26	0	\$700.00	tier - 2	tier - 3	R1013	26 December 1999	23
Id2323	1999	Dec	14	0	\$722.99	tier - 3	tier - 1	R1013	14 December 1999	23
Id2322	2002	Sep	19	0	\$750.00	tier - 3	tier - 1	R1012	19 September 2002	20
Id2321	1993	Aug	9	0	\$760.00	tier - 3	tier - 1	R1013	09 August 1993	29
Id2320	1996	Oct	22	0	\$760.00	tier - 3	tier - 3	R1013	22 October 1996	26
Id2319	1993	Jun	28	0	\$770.00	tier - 3	tier - 3	R1013	28 June 1993	29
Id2318	1996	Sep	18	0	\$770.38	tier - 3	tier - 2	R1012	18 September 1996	26
Id2317	1995	Dec	7	0	\$773.54	tier - 3	tier - 2	R1013	07 December 1995	27
Id2316	2004	Oct	7	0	\$830.52	tier - 3	tier - 2	R1011	07 October 2004	18
Id2315	2000	Nov	18	0	\$865.41	tier - 3	tier - 1	R1013	18 November 2000	22
Id2314	1993	Nov	27	0	\$896.21	tier - 3	tier - 1	R1013	27 November 1993	29
Id2313	1994	Oct	30	0	\$915.07	tier - 3	tier - 2	R1013	30 October 1994	28
Id2312	1995	Oct	28	0	\$928.59	tier - 3	tier - 1	R1013	28 October 1995	27
Id2311	2001	Aug	19	0	\$964.71	tier - 3	tier - 2	R1013	19 August 2001	21
Id2310	1994	Nov	22	0	\$1,006.65	tier - 3	tier - 2	R1013	22 November 1994	28
Id2309	2000	Oct	9	0	\$1,012.00	tier - 3	tier - 2	R1013	09 October 2000	22
Id2308	1999	Aug	5	0	\$1,019.00	tier - 3	tier - 2	R1013	05 August 1999	23
Id2307	1995	Oct	5	0	\$1,033.74	tier - 3	tier - 1	R1013	05 October 1995	27
Id2306	1994	Nov	17	0	\$1,044.00	tier - 3	tier - 2	R1013	17 November 1994	28
Id2305	2002	Jul	20	0	\$1,047.00	tier - 3	tier - 1	R1013	20 July 2002	20
Id2304	2002	Jul	1	0	\$1,049.00	tier - 3	tier - 2	R1012	01 July 2002	20
Id2303	1995	Jun	1	0	\$1,056.00	tier - 3	tier - 2	R1013	01 June 1995	28
Id2302	1993	Sep	28	0	\$1,068.00	tier - 3	tier - 1	R1013	28 September 1993	29
Id2301	2004	Aug	24	0	\$1,070.00	tier - 3	tier - 3	R1013	24 August 2004	18
Id2300	1992	Aug	13	0	\$1,071.00	tier - 3	tier - 3	R1013	13 August 1992	30

Data Transformation

- Split customer names into Title, First Name, and Last Name.
- Converted NumberOfMajorSurgeries into numeric values.
- Created Weight Status categories using BMI ranges.
- Created Diabetes Status categories using HbA1C ranges.
- Merged date, month, and year into Date of Birth
- Calculated Age based on dataset collection date (08-Jun-2023).
- Formatted healthcare charges as currency (\$).
- Merged all tables into a Healthcare master sheet using VLOOKUP

A	B	C	D
Customer Id	Title	First Name	Last Name
Id1	Ms.	Kelly	Hawks
Id2	Mr.	Matthew D	Lehner
Id3	Mr.	Phil	Lu
Id4	Ms.	Kelsey	Osborne
Id5	Ms.	Kristyn	Kadala
Id6	Mr.	Russell B.	Baker
Id7	Mr.	Scott	Macpherson
Id8	Mr.	Stephen	Hallman
Id9	Mr.	Patrick R.	Moran
Id10	Ms.	Brooke N.	Benner
Id11	Ms.	Paola Andrea	Fierro Vargas
Id12	Mr.	David	Franz
Id13	Mr.	Wade	Foster
Id14	Mr.	Franklin	Tenorio
Id15	Ms.	Leilani M.	Rios
Id16	Mr.	Philippe	Viau-Dupuis
Id17	Ms.	Jennifer A.	Cronin
Id18	Ms.	Christina M.	Noordstar
Id19	Mr.	Said Sr.	Boudalia
Id20	Mr.	John	Flor
Id21	Mr.	Myles	Fennon
Id22	Ms.	Madelyn C	Hribar
Id23	Ms.	Nicole	Tassello
Id24	Mr.	Eric A.	Mauricette
Id25	Mr.	Emiliano I.	Garcia
Id26	Mr.	Adam	Aioldi
Id27	Mr.	Zach	Cater-Cyker
Id28	Ms.	Stephanie W.	Hamm
Id29	Mr.	Stephen	Cox
Id30	Ms.	Natalie	Sachrajda
Id31	Ms.	Nicole J.	Gareri
Id32	Mr.	Jefferson D	Welch
Id33	Mr.	Brendan D.	Prindiville
Id34	Ms.	Courtney	Kincaid

Clipboard		Font		Alignment		Number		Styles							
G28															
=VLOOKUP(SA28,'Medical Examinations'!SA:\$I,6,FALSE)															
First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	smoker	Weight Status	Diabetes Status	Date of Birth	charges	Hospital tier	City tier	State ID	Age
Kelly	47.41	7.47	No	No	No		0 Yes	Obesity	Diabetes	12 October 1968	\$63,770.43	tier - 1	tier - 3	R1013	54
Matthew D	30.36	5.77	No	No	No		0 Yes	Obesity	Prediabetes	08 June 1977	\$62,592.87	tier - 2	tier - 3	R1013	46
Phil	34.485	11.87	Yes	No	No		2 Yes	Obesity	Diabetes	11 September 1970	\$60,021.40	tier - 1	tier - 1	R1012	52
Kelsey	38.095	6.05	No	No	No		0 Yes	Obesity	Prediabetes	06 June 1991	\$58,571.07	tier - 1	tier - 3	R1024	32
Kristyn	35.53	5.45	No	No	No		0 Yes	Obesity	Normal	19 June 1989	\$55,135.40	tier - 1	tier - 2	R1012	33
Russell B.	32.8	6.59	No	No	No		0 Yes	Obesity	Diabetes	04 August 1962	\$52,590.83	tier - 1	tier - 3	R1011	60
Scott	36.4	6.07	No	No	No		0 Yes	Obesity	Prediabetes	27 October 1994	\$51,194.56	tier - 1	tier - 3	R1011	28
Stephen	36.96	7.93	No	No	No		3 Yes	Obesity	Diabetes	27 June 1958	\$49,577.66	tier - 2	tier - 2	R1013	64
Patrick R.	41.14	9.58	Yes	No	Yes		1 Yes	Obesity	Diabetes	04 September 1963	\$48,970.25	tier - 1	tier - 2	R1013	59
Brooke N.	38.06	10.79	No	No	No		0 Yes	Obesity	Diabetes	29 December 1978	\$48,885.14	tier - 1	tier - 2	R1013	44
Paola Andrea	37.7	5.96	Yes	No	No		2 Yes	Obesity	Prediabetes	22 July 1959	\$48,824.45	tier - 2	tier - 1	R1011	63
David	42.13	11.9	No	No	No		0 Yes	Obesity	Diabetes	27 October 1965	\$48,675.52	tier - 1	tier - 2	R1013	57
Wade	40.92	8.41	No	No	No		0 Yes	Obesity	Diabetes	11 October 1962	\$48,673.56	tier - 1	tier - 2	R1013	60
Franklin	40.565	7.02	No	No	No		0 Yes	Obesity	Diabetes	01 December 1968	\$48,549.18	tier - 1	tier - 3	R1016	54
Leilani M.	36.385	7.59	Yes	No	No		2 Yes	Obesity	Diabetes	21 December 1961	\$48,517.56	tier - 1	tier - 3	R1024	61
Philippe	39.9	11.32	No	No	No		0 Yes	Obesity	Diabetes	27 August 1962	\$48,173.36	tier - 1	tier - 3	R1011	60
Jennifer A.	33.8	7.67	No	No	No		3 Yes	Obesity	Diabetes	16 November 1958	\$47,928.03	tier - 2	tier - 3	R1011	64
Christina M.	36.765	7.29	Yes	No	Yes		1 Yes	Obesity	Diabetes	05 August 1963	\$47,896.79	tier - 1	tier - 3	R1024	59
Said Sr.	36.955	4.72	Yes	No	No		1 Yes	Obesity	Normal	07 November 1964	\$47,496.49	tier - 1	tier - 3	R1012	58
John	42.9	11.41	No	No	No		0 Yes	Obesity	Diabetes	27 September 1971	\$47,462.89	tier - 1	tier - 2	R1013	51
Myles	36.3	11.5	Yes	No	No		2 Yes	Obesity	Diabetes	13 June 1961	\$47,403.88	tier - 1	tier - 3	R1011	61
Madelyn C	32.2	6.22	Yes	No	No		2 Yes	Obesity	Prediabetes	04 October 1959	\$47,305.31	tier - 2	tier - 1	R1011	63
Nicole	31.3	11.38	No	No	No		3 Yes	Obesity	Diabetes	03 September 1958	\$47,291.06	tier - 2	tier - 1	R1011	64
Eric A.	41.8	7.89	Yes	No	No		2 Yes	Obesity	Diabetes	28 December 1970	\$47,269.85	tier - 1	tier - 2	R1013	52
Emiliano I.	35.09	4.38	Yes	No	No		2 Yes	Obesity	Normal	14 July 1959	\$47,055.53	tier - 2	tier - 1	R1013	63
Adam	33.88	7.01	No	No	No		3 Yes	Obesity	Diabetes	15 September 1958	\$46,889.26	tier - 2	tier - 3	R1013	64
Zach	30.875	11.88	No	No	No		0 Yes	Obesity	Diabetes	17 November 1960	\$46,718.16	tier - 1	tier - 1	R1012	62
Stephanie W.	36.86	5.19	Yes	No	Yes		1 Yes	Obesity	Normal	23 June 1969	\$46,661.44	tier - 1	tier - 3	R1012	53
Stephen	35.86	6.74	Yes	No	No		2 Yes	Obesity	Diabetes	08 August 1961	\$46,599.11	tier - 1	tier - 1	R1013	61
Natalie	37.05	8.44	No	No	No		0 Yes	Obesity	Diabetes	06 December 1971	\$46,255.11	tier - 1	tier - 3	R1024	51
Nicole J.	43.89	8.71	No	No	No		0 Yes	Obesity	Diabetes	17 August 1978	\$46,200.99	tier - 1	tier - 1	R1013	44
Jefferson D	42.35	5.08	Yes	No	No		0 Yes	Obesity	Normal	02 December 1976	\$46,151.12	tier - 1	tier - 3	R1013	46
Brendan D.	31.35	6.86	No	No	No		0 Yes	Obesity	Diabetes	04 December 1962	\$46,130.53	tier - 1	tier - 1	R1012	60
Courtney	47.6	5.95	Yes	No	No		0 Yes	Obesity	Prediabetes	30 September 1985	\$46,113.51	tier - 2	tier - 3	R1011	37
Sydney L.	46.2	6.09	No	No	Yes		1 Yes	Obesity	Prediabetes	12 August 1979	\$45,863.21	tier - 2	tier - 1	R1013	43

Analysis & Visualizations

- Cancer history comparison among smokers and non-smokers using donut charts.
- Comparison of surgeries and HbA1C based on transplant history.
- Healthcare charges comparison across Weight Status and Diabetes Status using bar charts.
- Average hospital charges comparison across hospital tiers and states.
- Scatter plot analysis between Age vs BMI, Age vs HbA1C, and Age vs Charges.
- Created interactive slicers for Weight Status and Diabetes Status.

Dashboard KPIs

- Total Customers
- Average Healthcare Charges
- Average BMI
- Average HbA1C
- Total Smokers
- Diabetes Patients Percentage

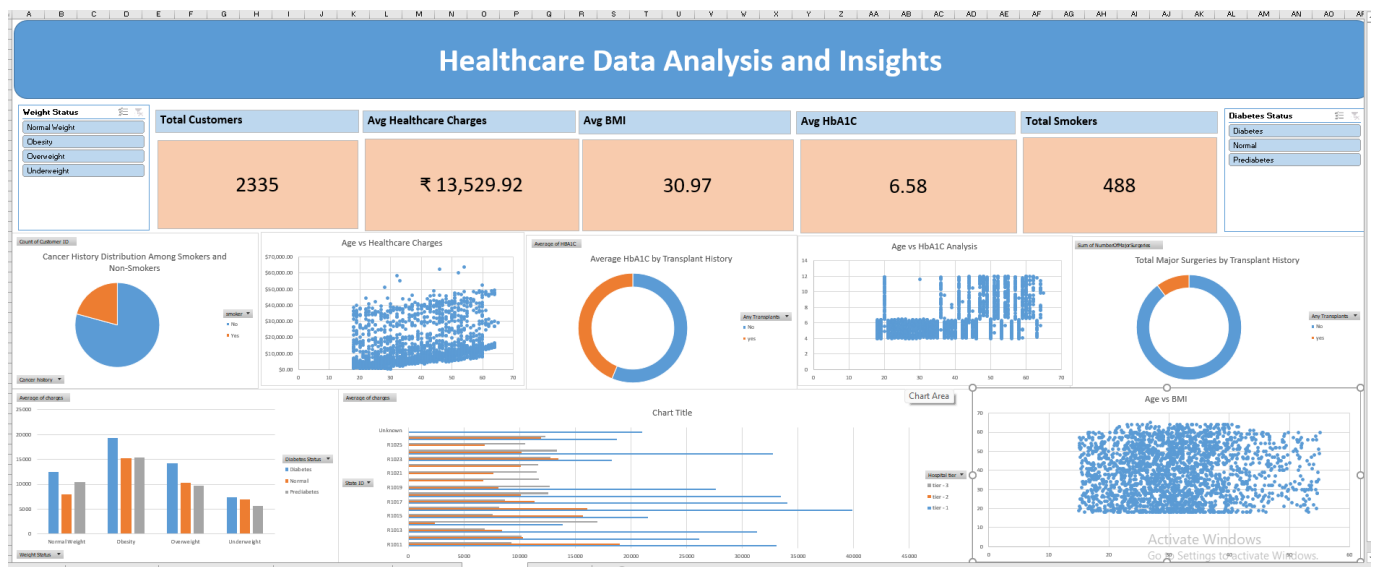
Dashboard Design Features

The dashboard was designed with multiple chart types, KPI cards, slicers, and interactive pivot charts to provide insights into patient health conditions and healthcare costs. Scatter plots were used for correlation analysis, while bar and donut charts were used for categorical comparisons.

Count of Customer I Column Labels			
Row Labels	No	Yes	Grand Total
No	1540	307	1847
Yes	404	84	488
Grand Total	1944	391	2335
Sum of NumberOfMajor			
Row Labels	Surgeries		
No	1417		
yes	162		
Grand Total	1579		
Average of HbA1C			
Row Labels			
No	6.67044272		
yes	5.187638889		
Grand Total	6.578997859		
Average of charges			
Row Labels	Diabetes	Normal	Prediabetes
Normal Weight	12447.61288	7938.505741	10492.13061
Obesity	19347.65735	15188.4366	15353.90224
Overweight	14269.01401	10327.15691	9737.940185
Underweight	7442.964923	6911.00561	5647.748276
Grand Total	16120.28721	12090.72298	12532.26268

Count of smoker	
Row Labels	
No	1847
Yes	488
Grand Total	2335
Count of Hospital tier	
Row Labels	
tier - 1	309
tier - 2	1340
tier - 3	694
Grand Total	2343
Count of City tier	
Row Labels	
tier - 1	734
tier - 2	813
tier - 3	796
Grand Total	2343
Average of BMI	
	30.97

Average of charges				
Row Labels	tier - 1	tier - 2	tier - 3	Grand Total
R1011	33081.37379	18997.28069	9250.361786	19466.24611
R1012	26111.45794	10334.27932	10179.94197	12016.47391
R1013	31328.10224	8439.2844	6890.756824	10514.9398
R1014	13891.86	2395.17	16950.485	13478.05692
R1015	21523.52	15675.54333	7570.755	13948.61
R1016	39868.61625	16075.9633	8133.00849	13589.26344
R1017	34070.59571	11365.62188	8667.366154	14806.10778
R1018	33475.82	10139.55333	12570.97	13272.78667
R1019	27621.61	8116.72	12706.976	13633.14346
R1020	6760.543333	11716.45333	9238.498333	
R1021	7668.178723	11518.78565	8933.378143	
R1022	10102.02667	11637.812	10650.52143	
R1023	18261.7575	13457.09458	12785.637	13786.14921
R1024	32732.18714	10174.346	13360.97889	12865.1495
R1025	6886.036842	10510.31381	8788.78225	
R1026	18709.798	11919.60786	12296.77973	12489.92107
Unknown	20996.53			20996.53
Grand Total	30129.19859	11865.26877	9462.269307	13529.91803



Conclusion

The Healthcare Analytics Dashboard successfully transformed raw healthcare data into meaningful business insights. The project demonstrates practical Excel skills including data cleaning, transformation, pivot tables, dashboarding, KPI creation, and interactive reporting.